WORKED SOLUTION

STEP I 1993, Question 2

Pure • Geometric Series

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/93-s1-q2

The key idea

Each series is a geometric progression with some terms removed. The exponents in $S_n(r)$ are the numbers that are *not* multiples of 3. The exponents in $T_n(r)$ are the numbers that are even *or* multiples of 3. So we write each sum as a combination of complete geometric progressions: we take a full progression and subtract the unwanted terms, or add two progressions and subtract the terms we have counted twice.

Solution

The sum of even powers

The left-hand side is a geometric progression with first term 1, common ratio r^2 and $n+1$ terms. Call it E. Then

$$E - r^2E = (1 + r^2 + \cdots + r^{2n}) - (r^2 + r^4 + \cdots + r^{2n+2}) = 1 - r^{2n+2},$$

since everything else cancels. If $|r| \neq 1$ then $r^2 \neq 1$, so we may divide by $1 - r^2$:

$$1 + r^2 + r^4 + \cdots + r^{2n} = \frac{1 - r^{2n+2}}{1 - r^2},$$

as required. (For real r, the condition $r^2 \neq 1$ is exactly $|r| \neq 1$.)

The series $S_n(r)$

The exponents $1, 2, 4, 5, 7, 8, 10, \ldots, 3n-1$ are the integers from 1 to $3n-1$ that are not multiples of 3. They come in pairs $3k+1,\ 3k+2$ for $k = 0, 1, \ldots, n-1$. So

$$S_n(r) = \sum_{k=0}^{n-1} (r^{3k+1} + r^{3k+2}) = (r + r^2) \sum_{k=0}^{n-1} r^{3k}.$$

The last sum is a geometric progression with ratio r^3 and n terms. For real r, $r \neq 1$ gives $r^3 \neq 1$, so

$$S_n(r) = \frac{r(1+r)(1-r^{3n})}{1-r^3}.$$

Equivalently, we can take all powers $r, r^2, \ldots, r^{3n}$ and remove the multiples of 3:

$$S_n(r) = \frac{r(1-r^{3n})}{1-r} - \frac{r^3(1-r^{3n})}{1-r^3}.$$

This second form is the one we need for the limit.

The limit. If $|r| < 1$ then $|r^{3n}| = |r|^{3n} \to 0$ as $n \to \infty$. Hence

$$S_n(r) \to \frac{r}{1-r} - \frac{r^3}{1-r^3}.$$

Now $\dfrac{r}{1-r} = \dfrac{1}{1-r} - 1$ and $\dfrac{r^3}{1-r^3} = \dfrac{1}{1-r^3} - 1$. The two -1s cancel, which leaves

$$S_n(r) \to \frac{1}{1-r} - \frac{1}{1-r^3},$$

as required.

The series $T_n(r)$

The exponents $0, 2, 3, 4, 6, 8, 9, 10, 12, \ldots, 6n$ are the integers from 0 to $6n$ that are even or divisible by 3 (or both).

> **Ask yourself** If we add the even powers to the powers that are multiples of 3, which terms have we counted twice?

An exponent is both even and a multiple of 3 exactly when it is a multiple of 6. So we use inclusion–exclusion: add the even powers, add the multiples of 3, and subtract the multiples of 6. Up to $6n$ there are

- even powers $r^0, r^2, \ldots, r^{6n}$, which form a GP with ratio r^2 and $3n+1$ terms;
- multiples of 3, $r^0, r^3, \ldots, r^{6n}$, which form a GP with ratio r^3 and $2n+1$ terms;
- multiples of 6, $r^0, r^6, \ldots, r^{6n}$, which form a GP with ratio r^6 and $n+1$ terms.

By the first result with n replaced by $3n$, together with the same argument for ratios r^3 and r^6, we get

$$T_n(r) = \frac{1-r^{6n+2}}{1-r^2} + \frac{1-r^{6n+3}}{1-r^3} - \frac{1-r^{6n+6}}{1-r^6}.$$

Every denominator is nonzero because $|r| \neq 1$. This is why the question needs $|r| \neq 1$ here and only $r \neq 1$ for S_n: $r = -1$ makes $1-r^2$ and $1-r^6$ vanish.

The limit. If $|r| < 1$, then r^{6n+2}, r^{6n+3} and r^{6n+6} all tend to 0. So

$$T_n(r) \to \frac{1}{1-r^2} + \frac{1}{1-r^3} - \frac{1}{1-r^6}.$$

This tidies up over the common denominator $1-r^6$, using $1-r^6=(1-r^2)(1+r^2+r^4)=(1-r^3)(1+r^3)$:

$$\lim_{n\to\infty} \mathrm{T}_n(r)=\frac{(1+r^2+r^4)+(1+r^3)-1}{1-r^6}=\frac{1+r^2+r^3+r^4}{1-r^6}.$$

The cases $r>1$, $r=1$, $r=-1$

We count the terms first. In each block of six exponents $6k, 6k+1, \ldots, 6k+5$, the ones that are even or divisible by 3 are $6k, 6k+2, 6k+3, 6k+4$, which is four of them. The blocks $k=0,\ldots,n-1$ cover the exponents up to $6n-1$, and the final term is r^{6n}. So $\mathrm{T}_n(r)$ has $4n+1$ terms.

- $r>1$**:** every term r^e satisfies $r^e\geq 1$, so $\mathrm{T}_n(r)\geq 4n+1$, and therefore $\mathrm{T}_n(r)\to\infty$.
- $r=1$**:** every term equals 1, so $\mathrm{T}_n(1)=4n+1\to\infty$. The formula breaks down here because its denominators are zero.
- $r=-1$**:** $(-1)^e$ is $+1$ for even e and -1 for odd e. In each block the exponents $6k, 6k+2, 6k+4$ are even and $6k+3$ is odd, so each block contributes $1+1-1+1=2$. Hence
$$\mathrm{T}_n(-1)=2n+1\to\infty.$$

In all three cases $\mathrm{T}_n(r)$ diverges to $+\infty$. Note that $r=-1$ also diverges: the odd multiples of 3 do not cancel enough of the even terms. Contrast this with a simple alternating series such as $1-1+1-\cdots$, which merely oscillates.

Checks

- Put $n=1$ in the S_n formula: $\frac{r(1+r)(1-r^3)}{1-r^3}=r+r^2$, which is correct.
- Put $r=\frac{1}{2}$ in the limit of T_n. The formula gives $\frac{1+\frac{1}{4}+\frac{1}{8}+\frac{1}{16}}{1-\frac{1}{64}}=\frac{92}{63}\approx 1.4603$, and adding several hundred terms of $\mathrm{T}_n(\frac{1}{2})$ directly gives the same value.
- As $r\to 1^-$, the limit of T_n behaves like $\frac{4}{6(1-r)}=\frac{2}{3(1-r)}$. This is consistent with four of every six powers being present, about two-thirds of the full series $\frac{1}{1-r}$.

Common slips

- Forgetting to subtract the multiples of 6, which counts $1, r^6, r^{12}, \ldots$ twice.
- Getting the number of terms wrong in each GP. There are $3n+1$ even exponents from 0 to $6n$, not $3n$.
- Assuming $\mathrm{T}_n(-1)$ oscillates like an alternating series, without actually adding up a block of terms.

Beyond STEP

Counting the terms in a union by adding the two sets and subtracting their intersection is the *inclusion–exclusion principle*. Combined with generating functions, it is a standard tool in combinatorics and number theory: the limit we found is the generating function of the non-negative integers that share a factor with 6. The same sieve idea, applied with every prime, gives Euler's product for the Riemann zeta function. Undergraduate analysis makes the step "$r^n \to 0$ when $|r| < 1$" rigorous and asks when infinite series may be rearranged in this way. Abbott's *Understanding Analysis* covers this well.

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